Machine Learning Foundations – Part 5

This final phase covers optimization techniques, gradient descent variants, logistic regression, and classification metrics with beginner-friendly explanations, light mathematics, and real-life examples.

1. What is Gradient Descent?

Gradient Descent is an optimization algorithm used to minimize a loss function by iteratively updating model parameters in the direction of the steepest decrease of the error. Real-life analogy: walking downhill to reach the lowest point of a valley.

2. Why Gradient Descent is Needed

Closed-form solutions are not feasible for large datasets or complex models. Gradient Descent provides an efficient and scalable way to learn parameters.

3. Learning Rate

Learning rate controls the step size during optimization. Too large causes overshooting, too small causes slow convergence.

4. Types of Gradient Descent

There are three main types: Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent.

5. Batch Gradient Descent

Uses the entire dataset to compute gradients. Stable but computationally expensive and slow.

6. Stochastic Gradient Descent (SGD)

Updates parameters using one data point at a time. Faster but noisy. Suitable for large datasets.

7. Mini-Batch Gradient Descent

Uses small batches of data. Balances speed and stability. Most commonly used in practice.

8. Effect of Learning Rate and Data

Learning rate and data distribution significantly impact convergence and final model quality.

9. Introduction to Logistic Regression

Logistic Regression is a classification algorithm used to predict probabilities of binary outcomes. Despite its name, it is a classification model.

10. Perceptron Problem & Solution

Perceptron fails for non-linearly separable data. Logistic regression solves this using a probabilistic approach.

11. Sigmoid Function

Sigmoid maps input values to probabilities between 0 and 1. It enables smooth decision boundaries.

12. Loss Function – Cross Entropy

Cross-entropy loss penalizes confident but wrong predictions heavily. It works well with sigmoid-based models.

13. Optimization using Gradient Descent

Logistic regression parameters are optimized using gradient descent to minimize cross-entropy loss.

14. Classification Metrics

Accuracy, Precision, Recall, F1-score, and Confusion Matrix evaluate classification models.

15. Problem with Accuracy

Accuracy fails for imbalanced datasets. Other metrics must be considered for fair evaluation.

Key Takeaways

Optimization drives learning. Logistic regression enables classification. Choosing the right metrics is crucial for real-world ML systems.